

Project Design Phase

Project Title: Enforcing Mandatory Fields Using UI Policies and Migrating Changes with Update Sets

1. Introduction

The project design phase is an important stage in the software development lifecycle where the system architecture and structure are defined. In this phase, the project requirements identified earlier are transformed into a detailed design that explains how the system will function. The purpose of the design phase is to provide a clear blueprint for developers so that the system can be implemented effectively.

For the project “**Enforcing Mandatory Fields Using UI Policies and Migrating Changes with Update Sets,**” the design phase focuses on creating a structured approach for implementing UI Policies within ServiceNow forms and managing configuration changes using Update Sets. The design ensures that mandatory fields are enforced dynamically and that system changes can be safely migrated between ServiceNow instances.

2. System Architecture

The system architecture for this project is based on the ServiceNow platform, which operates as a cloud-based service management system. The architecture consists of multiple ServiceNow instances, typically including:

- **Development Instance** – where configurations such as UI Policies are created and tested.
- **Testing Instance** – where configurations are verified for functionality and performance.
- **Production Instance** – the live environment where end users interact with the system.

The configurations developed in the development instance are captured using Update Sets and then migrated to higher environments to maintain system consistency.

3. Design of UI Policies

UI Policies are the core component of the project design. They control the behavior of form fields based on predefined conditions. The design includes the following elements:

Policy Conditions

Conditions define when the UI Policy should be applied. For example, when the category field in an incident form is set to a specific value, additional fields may become mandatory.

Policy Actions

Actions determine what happens when the conditions are met. Fields may be configured to become mandatory, visible, or read-only depending on the requirement.

Form Interaction

When a user interacts with the form, the UI Policy automatically checks the defined conditions and applies the necessary changes. This ensures that the user provides complete information before submitting the form.

4. Update Set Design

Update Sets are used to track and migrate configuration changes across ServiceNow environments. The design includes the following steps:

1. Create a new Update Set in the development instance.
2. Configure UI Policies and other system changes.
3. Capture these changes automatically in the Update Set.
4. Export or move the Update Set to another ServiceNow instance.
5. Preview and commit the Update Set in the target instance.

This structured process ensures that all system modifications are transferred accurately without missing components.

5. Data Flow Design

The data flow within the system is designed to ensure proper information collection and validation.

1. A user accesses a ServiceNow form such as the **Incident form**.
2. The user enters initial details about the issue.
3. Based on the selected category or condition, UI Policies activate automatically.
4. Additional fields become mandatory or visible.
5. The user completes all required fields before submitting the record.
6. The record is stored in the ServiceNow database with complete and accurate information.

This design ensures that the system collects the necessary data required for efficient incident management.

6. User Interface Design

The user interface design focuses on providing a simple and intuitive experience for users interacting with ServiceNow forms. UI Policies help guide users by clearly indicating required fields and preventing form submission when important information is missing.

Key UI design considerations include:

- Highlighting mandatory fields clearly.
- Ensuring that additional fields appear only when needed.
- Preventing users from submitting incomplete forms.
- Providing a clean and easy-to-use form layout.

This design approach improves the usability of the system and reduces user errors.

7. Security and Access Control

Security is an important aspect of the system design. Access to configure UI Policies and manage Update Sets is restricted to authorized administrators or developers. Role-based access control ensures that only permitted users can modify system configurations.

End users can only interact with the forms and submit records, while administrators manage the configuration and migration of system changes.

8. Testing Strategy

Before deploying the configurations to the production instance, the system design includes a testing phase. UI Policies are tested in the development environment to ensure that they behave correctly under different conditions.

After initial testing, the Update Set containing the configurations is migrated to the testing instance. Additional validation is performed to confirm that the system works correctly without affecting existing functionalities.

9. Conclusion

The project design phase provides a detailed framework for implementing the proposed solution. By designing UI Policies to enforce mandatory fields and using Update Sets to migrate configuration changes, the system ensures both data accuracy and configuration consistency within the ServiceNow platform.

The design created during this phase serves as a blueprint for the next stage of the project, which is the **project development phase**, where the planned configurations will be implemented and tested in the ServiceNow environment.